

CSC15S Assignment 2 – FERPA & VIM Pro

Total: 100pts (see due Date in Blackboard)

What to turn in: Upload the following to Blackboard Assignment2

(1) CSC15Slab2.docx (with all questions completed)

Part 1. Laws and Regulations

[30 pts. Estimated time to complete: 30 minutes]

Q1. Find federal regulations that govern the disclosure of software vulnerabilities. To start, think about the following scenario: Dr. Evil performed some fuzzy testing on Google chrome and found a vulnerability that can enforce the browser to download and execute malicious files. Can Dr. Evil sell this vulnerability online? State your opinion and please cite the related laws. Dr. Evil is very impatient with reading, please keep your response brief and clear. Please restrain your answer within 50 words. [30 pts]

You can't legally sell the vulnerability. Gramm-Leach-Bliley (1999) and Sarbanes-Oxley (2002) require companies to disclose vulnerabilities and secure systems, making selling them illegal.

Part 2. VIMTUTOR & VIMGenius!

[30 pts. Estimated time to complete: 1.5 hrs]

Our plan is to finish all the tutorials of [vimtutor](#) and finish the corresponding [VIMGenius](#). We suggest that you allocate a **consecutive chunk of 1.5 hrs** to complete all tutorials (just in case somehow VIMGenius lost your records – even though you tried to log in using the same email).

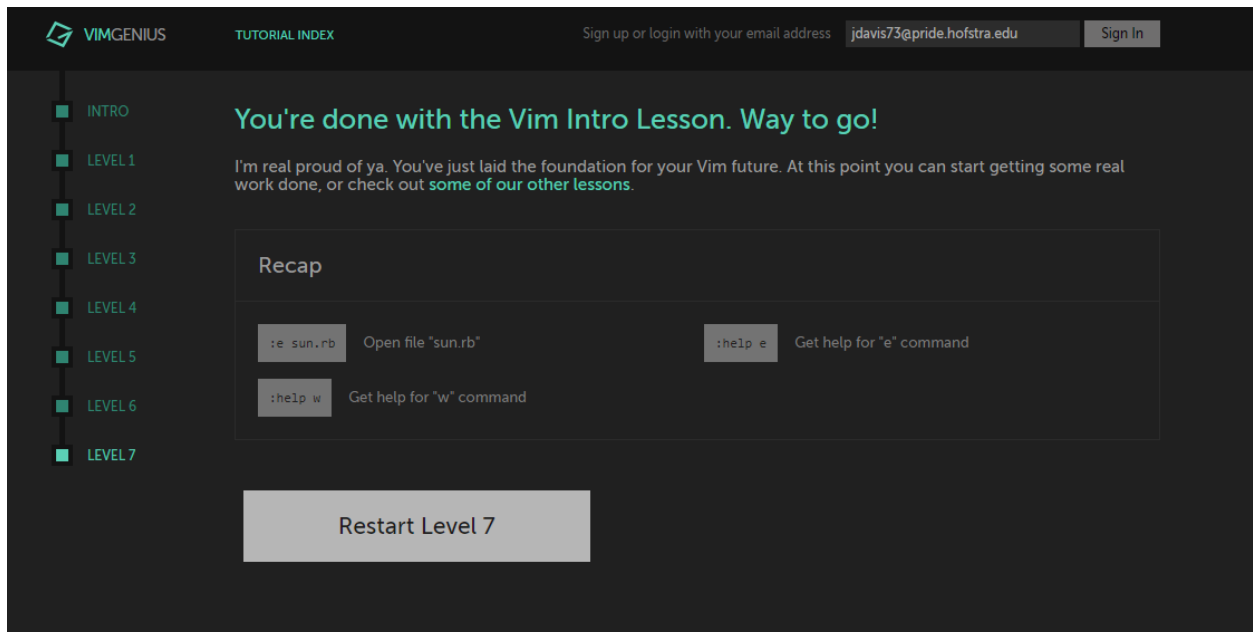
Please report the time efforts spent on each VIMTUTOR Lesson.

----- Worksheet -----

Q1. Please fill out the work log below. [10 pts]

Step	How much time did you spend? (in minutes)
1. VIMTUTOR Lesson 3	16
2. VIMTUTOR Lesson 4	13
3. VIMTUTOR Lesson 5	14
4. VIMTUTOR Lesson 6	22
5. VIMTUTOR Lesson 7	9
Total	72

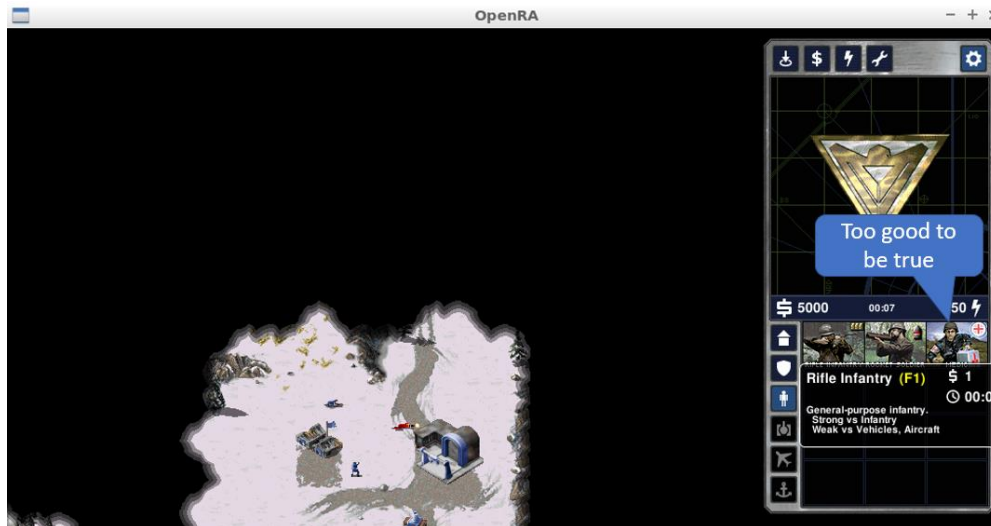
Q2. Please paste the **success screen** of your **VIMGenius Level 7** as shown below. [20 pts]



Part 3. Let's Cheat ... Red Alert

[40 pts. Estimated time to complete: 40 minutes]

Dr. Evil used to be a shameless cheater in gaming. Using a binary debugger (like gamebuster), he used to lock up the memory of games so that his player never dies. You will do something similar ... Observe the following screenshot of OpenRA (open source re-implementation of Red Alert, one of the most frequently played instant strategy games/ when Dr. Evil was a sophomore). Did you smell something strange?



Your goal in this mission is to **modify the source code of OpenRA** so that it only costs **one dollar** to create any soldier! Dr. Evil has already done most of the job. But unfortunately, there are certain syntax errors introduced. Please follow the steps below to fix source bugs. First, you have to navigate to the source code tree. The following is the command to type in a terminal window.

```
cd ~/Desktop/csc15s/openra/OpenRA
```

Here, the “~” is a shortcut to your homefolder (/home/csc15s). The “OpenRA” directory contains all the source code of red alert. Let’s first patch the Makefile of the project a bit. Create a file called **update.sh** with the following two lines:

```
sed -i "s/geoip-dependencies:/nouse/" Makefile
sed -i "s/geoip-dependencies/ /" Makefile
```

To run update.sh, we have to change its permission bits first: “**chmod 755 update.sh**” and then run it as “**./update.sh**”. Now To compile the project, you can type “**make all**” in the terminal, and you will find tons of syntax errors listed. Each with a **line number**, as shown in the following Figure. Your job is to fix them all (only best effort is needed as many of you don’t have experiences with the C# programming language yet). But after you have fixed one bug, if you run “make all” again in the terminal you will see one less bug out there.

Error is at Line 31
Column 36

Detailed Error Message

```
csc15s@csc15s-VirtualBox ~/Desktop/csc15s/openra/OpenRA$ make
CSC OpenRA.Game.exe
OpenRA.Game/WAngle.cs(31,36): error CS1519: Unexpected symbol 'Zero' in class, struct, or interface member declaration
OpenRA.Game/WAngle.cs(31,38): error CS1519: Unexpected symbol '=' in class, struct, or interface member declaration
OpenRA.Game/WAngle.cs(31,50): error CS1525: Unexpected symbol '0'
OpenRA.Game/WAngle.cs(31,44): error CS0106: The modifier 'new' is not valid
```

In the following table, please fill out the VIM commands that you have used in fixing each bug. You should try to use **the shortest command sequence** as possible, which helps you to achieve the maximum editing speed. As an example, we have listed the VIM commands (and the keystrokes) for fixing bug B1 (note the commands are separated by “,”).

ID	File	Line No	How to Fix	Your VIM Command
B1	OpenRA.Game/WAngle.cs	31	Change reAdOnLy to readonly	31gg, 2w, cw, readonly
B2	OpenRA.Game/WAngle.cs	19	Change the e in IEquatable to a (that is: IEquatable)	19,fl,fa, r, a,
B3	OpenRA.Game/WAngle.cs	34	Change oPeRator to operator	34, fo, cw, operator
B4	OpenRA.Game/WAngle.cs	41	There are two) after GetHashCode . Drop the extra one.	41,%,x
B5	OpenRA.Game/WAngle.cs	53	Comment out lines 53 to 55. That is: insert // at the beginning of each line (note: for C# the comment is NOT #). We suggest doing a string replacement: replace ^ (beginning of line) with \\ (note: the reason for \\ is because / is a special symbol in vim and it needs to be escaped)	53,55s/^\\
B6	OpenRA.Game/Player.cs	50	Append c after the publi . Note: to move to the end of the word, use the e command and then a to append.	50,fi,a,c
B7	OpenRA.Game/Player.cs	61/58	The error at line 61 is actually caused by line 58. Append ; at the end of line 50 (after BotType)	58,A,;
B8	OpenRA.Game/Player.cs	109	You need to remove 3 extra copies of FactionInfo .	/FactionInform , d3w
B9	OpenRA.Game/FieldLoader.cs	691	Function BuildTypeInfo is missing its function body. Copy the source code from fieldloader.txt , and paste it inside the function body. Hint: use :split fieldloader.txt to open a second window and then use ctrl+w j/k to navigate between the windows. Perform copy and paste.	691,dd,:split fieldloader.txt, cntrl+w,VG, y, cntrl+w, cntrl o, p
B10	OpenRA.Game/Actor.cs	49	You only need one copy of “public int Generation” . Delete the rest. (drop lines 50 to 59)	:50,59d
B11	OpenRA.Game/Actor.cs	56 and seven other lines	Replace all Disposedd with Disposed in the entire file.	:%s/Disposedd/Disposed/g
B12	OpenRA.Game/Actor.cs	88 and many other lines	Replace all Actorid with ActorID for the entire file	:%s/Actorid/ActorID/g
B13	OpenRA.Game/Download.cs	59	Paste the body from download.txt (similar to B9, using :split)	59, :split download.txt, cntrl w, VG, y, cntrl w,p
B14	mods/ra/rules/infantry.yaml	Multiple lines	Replace any line which looks like “Cost: some number” with “Cost: 1” . You can use a regular expression Cost: .* as the search pattern. Here “.*” means any arbitrary string (wild card)	:%s/Cost: .*/Cost: 1/g

Now let's make it work!

1. Start the game by `./run.sh` in terminal
2. Click **"SinglePlayer"**
3. Select **"Missions"**
4. Select Mission **"04: Ten to One"**

5. On the right top weapon palette, click the **"Personnel"** (3rd tab in the build menu) and show me the **\$1 cost** of creating soldiers. Please paste the **screenshot** below to prove your success. [Hint: if it's too hard to capture the \$1 sign, **simply create ONE soldier** and **show that you have \$4999 left**] [10 pts]

